

Benjamin Kaminow

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Research Interests

My current research focuses on evaluating structure-based graph neural networks for protein-ligand binding affinity prediction in the context of rational small molecule drug design. Specifically, I am interested in exploring why structure-based models underperform relative to ligand-only models, with an emphasis on physical biases such as E(3)-invariance and -equivariance. More generally, I am excited by any research that combines structural biology, physics, and machine learning.

Education

Ph.D. in Computational Biology and Medicine

May 2021 – Sept 2026 (Scheduled)

Memorial Sloan Kettering Cancer Center
and Weill Cornell Medical College, New York, NY

Advisor: John Chodera

Thesis: “Structure-based graph machine learning methods for protein-ligand binding affinity prediction”

B.S. in Chemical and Biomolecular Engineering

2015 – 2019

The Johns Hopkins University, Baltimore, MD

Research Experience

The Chodera Lab, Memorial Sloan Kettering Cancer Center

Jan 2022 – Present

Graduate Student

- Investigating the importance of incorporating physical biases in structure-based machine learning models for protein-ligand binding affinity prediction
- Assessing the performance, data efficiency, and generalizability of structure-based graph neural networks in the task of predicting protein-ligand binding affinities
- Developing a framework for modularly adapting arbitrary neural network architectures for structure-based machine learning
- Supported the drug discovery efforts of the AI-driven Structure-enabled Antiviral Platform through development of a computational chemistry software suite designed to assist at key stages of the drug discovery pipeline
- Conceptualized and planning appropriately scoped projects for rotation graduate students and summer undergraduate interns, and mentoring these students during their time in the lab
- Implemented an alternative atom enumeration scheme in the **espaloma** molecular machine learning model

The Dobin Lab, Cold Spring Harbor Laboratory

June 2019 – June 2021

Computational Science Developer

- Evaluated functional effects of replacing the current reference genome with different consensus genomes using the 1000 Genomes Project
- Analyzed scRNA-seq data from Tabula Muris to find differential splicing usage
- Developed ConsDB, a Python package for facilitating the use of large databases of genome variation
- Analyzed ChIP-seq data from different maize tissues as a part of the MaizeCODE project, in order to better understand maize epigenetics

The Karchin Lab, Institute for Computational Medicine

June 2018 – May 2019

Undergraduate Research Assistant

- Developed an updated method of finding similar alleles to increase the accuracy of the MHC-nuggets predictor on rare MHC alleles
- Explored and analyzed protein structure files to find new, biologically relevant features to add to the MHCnuggets machine learning model
- Generated predicted IC50 values using Python and Bash scripting to run the MHCnuggets pipeline
- Analyzed generated IC50 data using Python and R to perform Mann-Whitney U tests and generate Kaplan-Meier curves

The Betenbaugh Lab,

Jan 2017 – Sept 2018

Advanced Mammalian Biomanufacturing Innovation Center

Undergraduate Research Assistant

- Developed an integrated platform to combine Flux Balance Analysis and glycosylation modeling technologies by writing a MATLAB script to convert FBA results into the appropriate format, and interface between MATLAB and Excel
- Performed *in silico* optimization experiments on Chinese hamster ovary cells using the COBRA Toolbox for MATLAB to improve cell culture growth medium
- Utilized the COBRApy toolbox and high-performance computing to perform sampling experiments on Genome Scale Models in order to increase understanding of cell amino acid needs

The Schulman Lab, Institute for NanoBioTechnology

March 2016 – Jan 2017

Undergraduate Research Assistant

- Quantitated and analyzed data from experiments pertaining to the degree of swelling of DNA-crosslinked hydrogels
- Designed experiments to measure and quantitate the extent of polymerization of DNA-crosslinked hydrogels in order to better understand the chemical properties of the system

Publications

Journal Articles

Kaminow, B., Payne, A. M., MacDermott-Opeskin, H., Singh, S., Chodera, J. D., “Physical biases improve performance of structure-based binding affinity models,” *Draft available upon request*, 2026.

Payne, A. M., **Kaminow, B.**, MacDermott-Opeskin, H., Pulido, I., Scheen, J., Castellanos, M. A., Fearon, D., Chodera, J. D., Singh, S., “How many crystal structures do you need to trust your docking results?,” *J Med Chem*, *in press*, 2026.

MacDermott-Opeskin, H., Scheen, J., Wognum, C., ... **Kaminow, B.**, ... Chodera, J. D., “A Computational Community Blind Challenge on Pan-Coronavirus Drug Discovery Data,” *J Chem Inf Model*, 2026.

López-Ríos de Castro, R., Rodríguez-Guerra, J., Schaller, D., Kimber, T. B., Taylor, C., White, J. B., Backenköhler, M., Groß, J., Payne, A. M., **Kaminow, B.**, Pulido, I., Singh, S., Kramer, P. L., Pérez-Hernández, G., Volkamer, A., Chodera, J. D., “The Journey of Data: Lessons Learned in Modeling Kinase Affinity, Selectivity, and Resistance [Article v1.0],” *Living J Comput Mol Sci*, 2025.

Boby, M. L., Fearon, D., Ferla, M., Mihajlo, F., Koekemoer, L., Robinson, M. C., **The COVID Moonshot Consortium**, Chodera, J. D., Lee, A., London, N., von Delft, A., von Delft, F., “Open science discovery of potent noncovalent SARS-CoV-2 main protease inhibitors,” *Science*, 2023.

Wang, Y., Fass, J., **Kaminow, B.**, Herr, J. E., Rufa, D., Zhang, I., Pulido, I., Henry, M., Bruce Macdonald, H. E., Takaba, K., Chodera, J. D., “End-to-end differentiable construction of Molecular Mechanics Force Fields,” *Chem Sci*, 2022.

Kaminow, B., Ballouz, S., Gillis, J., Dobin, A., “Pan-human consensus genome significantly improves the accuracy of RNA-seq analyses,” *Genome Res*, 2022.

Shao, X. M., Bhattacharya, R., Huang, J., Sivakumar, I. A., Tokheim, C., Zheng, L., Hirsch, D., **Kaminow, B.**, Omdahl, A., Bonsack, M., Riemer, A., Velculescu, V., Anagnostou, V., Pagel, K.,

Karchin, R., “Prediction of MHC Class I and II Neoantigens with MHCnuggets,” *Cancer Immunol Res*, 2019.

Conference Papers

Kaminow, B., “Model construction for structure-based ML for binding affinity predictions,” *Computer Aided Drug Design Gordon Research Symposium [Oral Presentation]*, 2025.

Awards and Honors

- NSF Graduate Research Program, 2023
- Dean’s List, Fall 2015-Spring 2016, Spring 2017-Spring 2019
- Graduated from Johns Hopkins University with General Honors and Departmental Honors, 2019

Teaching

Foundations of Computational Biology and Bioinformatics *Spring 2019*

Undergraduate Teaching Assistant

- Assisted in planning of lectures and practical assignments
- Led weekly two-hour practical programming sessions for around 40 undergraduate and graduate students
- Mentored students through an original final project
- Automated the process of grading homework assignments by creating Bash scripts

Peer Led Team-Learning Academic Support Group (PILOT) *Fall 2017 – Spring 2019*

Physics I Head PILOT Leader

- Acted as a liaison between other Physics I PILOT Leaders and our supervisor
- Facilitated meetings with 5-10 other Physics I PILOT Leaders once a week for two hours
- Led problem-solving sessions for 10-15 undergraduate students in Physics I one-two times a week for two hours each

Skills

Machine Learning/Deep Learning: Hyperparameter tuning, model architecture selection, pre-training, data processing, graph neural networks, PyTorch, PyTorch Geometric, Weights & Biases, scikit-learn

Computational Chemistry: Structure visualization, molecular docking, OpenEye Toolkits, RDKit, PyMOL

Scientific Computing: High-performance computing, Git, Linux, data visualization, NumPy, Matplotlib, Seaborn, SciPy

Programming Languages: Python, Bash, R, Rust, C++